

03-11 Lecture: Blockchain & Cryptocurrencies Course – UFSC

Introduction, Structure, and Foundational Concepts

Date & Time: 2026-03-11 18:47:15

Location: Florianópolis, Brazil (UFSC – Universidade Federal de Santa Catarina)

Instructor: Jean Martina

Theme

This introductory lecture series for the Blockchain and Cryptocurrencies course at UFSC establishes course structure, objectives, grading philosophy, and foundational context. Students will learn the core building blocks of blockchain technology — rooted in decades-old cryptographic and data-structure concepts — and apply that knowledge to conceive and prototype a crypto startup, embracing AI tools and an innovation-first mindset throughout. The course weaves together the instructor's personal research background, the technical curriculum, the vibrant real-world Web3 ecosystem emerging in Florianópolis, and a project-oriented evaluation methodology emphasizing continuous partial deliveries over a single high-stakes exam.

Key Points

1. The course is optional and open to students from any degree program at UFSC, deliberately removing prerequisites to foster interdisciplinary teams capable of identifying real-world problems that blockchain can solve.
2. The primary goal has evolved from deep technical implementation to an innovation track: students will learn enough blockchain fundamentals to design and pitch a crypto startup as an MVP by semester's end, culminating in a business canvas, white paper, and pitch deck.
3. Lecture notes are auto-generated from audio recordings so that class time can be used for discussion rather than note-taking.
4. The course runs 18 weeks (~3 hours of content per week) and includes guest speakers from major blockchain ecosystems (Ethereum, Solana, Polkadot,

- Chainlink, Starknet) as well as practitioners from local innovation initiatives.
5. None of the underlying technology in blockchain is truly new: data structures date to the 1950s, hashes to the 1960s, public-key cryptography to the 1970s, and peer-to-peer networking to 1994. The genuine innovation is the game-theoretic “glue” binding these elements together.
 6. Blockchain’s three reinforcing pillars are: **History/Integrity, Rules, and Value/Incentives** — if any one pillar weakens, the whole system shakes.
 7. A cryptocurrency “wallet” is essentially a public key — understanding basic cryptography is necessary to demystify common blockchain terminology.
 8. The course will actively embrace AI-assisted (“vibe”) coding rather than restrict it, equipping students with tools to deliver working prototypes regardless of their coding background.
 9. The semester grade is composed of a theoretical exam and a series of practical partial deliveries — not a single final project submission — with grades auto-distributed equally to all group members.
 10. The instructor has deep personal history with the field: a PhD (2006–2010) in formal verification of cryptographic protocols, early exposure to Bitcoin’s whitepaper in December 2008, and even mining 50 BTC — which were later lost/abandoned.
 11. E-money (e.g., David Chaum’s blind signatures) predates cryptocurrencies and offered strong anonymity but relied on a trusted central entity; cryptocurrencies replaced that trust with decentralization.
 12. Brazil’s CBDC experiment, Drex, was officially abandoned by the Brazilian Central Bank due to fundamental incompatibilities: lack of transaction privacy, inability to control wallet balances, and difficulty of regulation — all by design in public blockchains.
 13. BRICS Pay was originally conceived in November 2019 as a cryptocurrency designed to bypass SWIFT and US financial dominance; it is now moving forward in a modified form.
 14. Florianópolis has become one of the most active Web3 hubs in Brazil and Latin America, hosting foundations and teams from Solana, Chainlink, Polkadot, Transfero, and others — many of whom will guest-lecture in this course.
 15. The course participates in SINOVA (CInova), UFSC’s university innovation competition; two selected teams represent the course, and winning earns a full extra point on the final grade.
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Highlights

- "There is no new technology; it's just how you put the things together. It was made in a clever way because one thing reinforces the other." --

Jean Martina

- "Either you follow the protocol or you gain nothing. And if you try to cheat the protocol, you lose everything." -- Jean Martina
- "A historian in the future may have to tell history by looking into a blockchain – and when looking into a blockchain, he knows that could not be changed; whatever was written in there is what actually happened at that point." -- Jean Martina
- "A wallet in cryptocurrency is not this. A wallet is a public key." -- Jean Martina
- "Most of the professors are probably thinking of a way to fight the usage of GPT when you are coding – we'll do the opposite." -- Jean Martina
- "This is not the standard mandatory course. We are here to inform you of new things that are happening in this domain – so the lectures will be much more easygoing." -- Jean Martina
- "The technology was created to be against central banks. It's a philosophical problem with banks." -- Jean Martina
- "The difference of a smart contract to that rental agreement is that the smart contract is written in code. And by being written in code, its interpretation is deterministic." -- Jean Martina
- "Identification doesn't mean de-anonymization. We can do both things at the same time." -- Jean Martina
- "Hyperinflation drives people to safety." -- Jean Martina
- "Whoever created Bitcoin got the first page of a newspaper in 2008 complaining about the central banks – and that message is embedded in Bitcoin; it can never be deleted." -- Jean Martina
- "Ledger is a technology well known to humankind – invented even before money. Distributed ledger is just a distributed computing version of that six-thousand-year-old technology." -- Jean Martina
- "I think it is unfair having only one exam, because if there is something wrong with any one of you, then the semester is busted." -- Jean Martina
- "I don't prepare very much for the lectures because there is no preparation for the lifeness of the interaction." -- Jean Martina
- "We end up learning as a group." -- Jean Martina
- "Choose wisely. Because like, life is like that – you always have to carry people around. How do you avoid that? Choose wisely." -- Jean Martina

- "Either you give up or you pass." -- Jean Martina
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Topics

1. Course Administration & Structure

The course is a 72-lecture-hour (50 min each), 18-week program conducted entirely in English despite being registered in Portuguese in the UFSC system. Attendance flexibility is built in for team-specific mentoring sessions, and lecture notes are auto-produced from audio so students can focus on discussion. The schedule is published on Moodle and updated continuously due to travel, strikes, power outages, and other disruptions.

- Classes are audio-recorded and transcribed into detailed lecture notes (last semester's produced 38 pages), eliminating the need to take notes during class.
 - The course is officially listed in Portuguese due to UFSC system limitations, but a clause in the methodology section confirms it is run in English; English is also maintained for inclusivity toward international students arriving from France, Norway, Germany, Portugal, and Latin America.
 - Effective content time is approximately 3 hours per week (15 minutes dropped at the start and end of each session); classes typically end around 9:00–9:30 PM rather than the scheduled 10:00 PM cutoff.
 - A second professor (Custódio) is formally listed for bureaucratic/coverage purposes but will not actively co-teach.
 - Some weeks will have non-mandatory attendance when team-specific crypto startup mentoring sessions replace full-group lectures.
 - The schedule is already on Moodle and will be continuously updated; uncontrollable disruptions include rats chewing electrical cables causing power outages, Brazil World Cup game Wednesdays, bus strikes, and campus flooding.
 - The instructor travels extensively — 32 trips in one year, nine of them to Europe — with key scheduled absences including Barcelona (April 15), Bogotá (May 13), and Berlin or Luxembourg (early June). When away, pre-recorded video lectures are released; the following in-person session begins with a Q&A before the deliverable deadline.
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2. Course Objectives & Innovation Track

The course has deliberately shifted from a deep-tech computer science elective to an innovation-focused program whose central deliverable is a crypto startup

MVP, making it accessible and relevant to students from any discipline.

- The original prerequisite (data structures + computer networks) was dropped to open the course to all degrees and foster interdisciplinary problem-solving.
 - The main objective is to learn enough blockchain technology to create something new, structured as a crypto startup proposition.
 - Students will work in teams; the instructor holds tailored mentoring sessions with each team rather than generic group feedback.
 - Even students without strong coding skills will be equipped to deliver an MVP through AI-assisted (“vibe”) coding tools introduced during the semester.
 - The course sits on UFSC’s innovation track, distinguishing it from standard mandatory technical courses.
 - The course is shifting from a technology-deep outline to a curiosity-based approach; the instructor regards leveling a class with widely varying prior knowledge as the hardest pedagogical challenge, and deliberately keeps early lectures as “teasers” rather than deep dives.
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3. Interdisciplinary & Open Enrollment

By removing prerequisites and opening enrollment to all UFSC programs, the course intentionally attracts students with diverse backgrounds — from history and economics to biomedical and mechanical engineering — because the most interesting blockchain applications emerge from non-CS problem domains.

- Past participants have included students from history, economics, international relations, information science, biomedical, electronic, and mechanical engineering.
 - A history student coined the concept of the “historian of the future” — someone who reads blockchain records as immutable primary sources of what actually occurred.
 - Blockchain may serve as a means to solve problems in virtually any field, not just finance or computer science.
 - Having some programming familiarity or prior cryptography knowledge accelerates understanding but is not required.
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4. Foundational Technology: Nothing Here Is New

All the building blocks of blockchain predate Bitcoin by decades; the true innovation of 2008–2009 was the game-theoretic protocol design that stitched those components together into a self-reinforcing system where cheating is economically irrational.

- Data structures (linked lists, pointers): 1950s.
 - Hashes as indexes and integrity controls: 1960s.
 - Public-key cryptography: publicly known from 1977 (RSA); later revealed that British intelligence used it as early as 1972.
 - Peer-to-peer networking: 1994.
 - Commitments (hash-and-reveal schemes): 1980s.
 - Homomorphic encryption: ~1997–1998; zero-knowledge proofs: ~2004.
 - The genuine novelty of Bitcoin/blockchain is the game-theoretic glue: the protocol design ensures that following the rules is the dominant strategy and cheating results in total loss.
 - Technically, a blockchain is a linked list where pointers are replaced by hash pointers, providing integrity control over each successive element.
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5. Three Pillars of Cryptocurrencies

Blockchain-based systems are stable when three mutually reinforcing pillars — History/Integrity, Rules, and Value/Incentives — are all functioning; weakness in any one pillar destabilizes the whole ecosystem.

- **History/Integrity:** The chain's immutability ensures that recorded events cannot be altered retroactively.
 - **Rules:** Protocol rules (consensus mechanisms, smart contract logic) govern what can and cannot happen on the chain.
 - **Value/Incentives:** Participants must have an economic or systemic reason to follow the rules; incentive engineering is especially challenging for non-financial applications.
 - For non-financial applications, designing clear incentives is significantly harder than for currency-based systems and is a core topic of the course.
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6. Cryptographic Primitives & Key Concepts

Students will build up from classical primitives (hashes, public-key cryptography) through modern constructs (zero-knowledge proofs, homomorphic encryption, commitments) to understand why blockchain properties such as immutability, privacy, and trustless execution actually hold.

- A **wallet** = a public key; understanding public-key cryptography is essential to demystify basic cryptocurrency terminology.
- **Hash pointers** replace standard pointers in the blockchain data structure, enabling integrity verification.

- **Commitments:** publish a hash now; reveal the pre-image later — second-preimage resistance guarantees the commitment is binding.
 - **Zero-knowledge proofs (ZKPs):** allow one party to prove knowledge of information without revealing the information itself; foundational to modern privacy-preserving blockchain protocols.
 - **Homomorphic encryption:** enables computation on encrypted data — increasingly relevant to blockchain applications.
 - David Chaum invented **blind signatures**: signing a document inside an envelope — you sign something you never see — enabling anonymous digital money and electronic elections.
 - The course will also cover consensus mechanisms: Proof of Work, Proof of Stake, and Byzantine Agreement protocols (over 50 catalogued types exist).
 - Cryptographic identification of users does not require de-anonymization — privacy and identity can coexist.
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7. Instructor Background & Early Blockchain Encounters

Jean Martina's PhD (2006–2010) in formal verification of cryptographic protocols led him to work on e-sealed-bid auctions using e-money (David Chaum's blind signatures). He read the Bitcoin whitepaper in December 2008 and ran the software in January 2009 with mixed feelings. He later mined 50 BTC during a postdoc, only to realize their value much later — a missed fortune he uses as a self-deprecating teaching moment.

- PhD focused on formal verification of cryptographic protocols, specifically electronic sealed-bid auction protocols using e-money commitments.
 - Bitcoin whitepaper was published November 2008; instructor read it in December 2008 and ran the software in January 2009.
 - He mined one block (50 BTC) during a 2012 postdoc at UBC; by the time a student told him "1 Bitcoin = \$100," the coins were gone/inaccessible.
 - Key distinction: e-money required trust in a central bank to issue currency but offered strong anonymity; cryptocurrencies replaced central trust with decentralization.
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8. Course Technical Curriculum

The course systematically builds from cryptographic primitives through blockchain building blocks, distributed ledger theory, consensus mechanisms, transaction models, and smart contract development — ensuring students share a common technical vocabulary before exploring applications.

- **Week 1:** Cryptographic parameters necessary for subsequent discussions.
 - **Weeks 2–3:** Concepts — assets, wallets, transactions, mining, nodes, distributed consensus.
 - **Weeks 4–6:** Blockchain properties (expanded from 2 to 3 weeks this semester).
 - **Platforms and ecosystems:** guest speakers from Solana, Chainlink, Polkadot, and other foundations rather than instructor-led explanation.
 - **Applications:** cryptocurrency (Bitcoin, altcoins), stablecoins, DeFi (guest lecture from a local Web3 firm — also a recruiting opportunity).
 - **Smart contract coding:** one lecture on Solidity basics and one on Hyperledger Fabric chaincode in Golang, both led by Lucas (master’s student).
 - **Additional planned lectures:** verifiable credentials, DeFi.
 - **Final three lectures:** business canvas creation, white paper writing, pitch deck construction — the course’s main deliverables.
 - **Newer cryptographic primitives** — including zero-knowledge proofs, homomorphic encryption, and commitments — will be covered beyond the classic Bitcoin-era hash-and-public-key foundation.
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9. Florianópolis as a Web3 Hub & Guest Lecture Ecosystem

Florianópolis has emerged as one of Brazil’s and Latin America’s leading Web3 hubs, rivaling São Paulo relative to its size and now competing internationally with Vietnam. The instructor leverages this local ecosystem to bring foundation representatives directly into the classroom, connecting students to hackathon opportunities, funding leads, and potential employment.

- Florianópolis now attracts major Web3 investment previously concentrated in São Paulo and Rio; crypto professionals prefer high-quality-of-life locations.
- Physical Web3 infrastructure in the city: a German hub at Primavera Garden, a founder house in Jurerê hosting Solana/Chainlink/Polkadot teams, a new crypto hub above Giassi in Sequentzium, and Transfero’s local presence.
- Kawea from Super Team Solana visited last semester; 4–5 student projects were built on Solana; at least one team is still active and connected to Solana’s global hackathon (prize: ~\$200,000).
- Guest speakers provide: technology explanation, real application scenarios, financing leads, coding guidance, and direct access to the worldwide hackathon ecosystem.
- The instructor rotates guest speakers each semester to give all ecosystem players fair visibility.
- Guest lectures also include practitioners from IP Village, a local initiative using DLT to build a “city of the future” in Florianópolis.

10. Blockchain Ecosystems & Non-Financial Applications

Beyond Bitcoin and Ethereum, a range of ecosystems (Solana, Polkadot, Chainlink, Starknet) each take different design tradeoffs; students will learn to distinguish financial from non-financial blockchain use cases and select appropriate technology accordingly.

- Distributed Ledger Technologies (DLTs) are not always chain-structured; some use graph-based data structures, and the right choice depends on the application.
- Non-financial applications include: Web3 decentralized apps, supply-chain tracking, tokenization of physical assets, and governance systems.
- **Smart contracts** are the programming model for deploying applications on top of a blockchain; their “smartness” lies in trustless, deterministic execution enforced by the chain itself — written in code, their interpretation is conflict-free before execution begins.
- A guest session on **vibe coding** (AI-assisted development) is scheduled for April 1st to demonstrate how students can build blockchain applications without deep coding expertise.

11. Drex (Brazil's CBDC) — Rise, Fall, and Implications

Brazil's Central Bank Digital Currency, Drex, was officially discontinued on August 5th of the prior year at Blockchain in Rio, announced by Gabriel Galipolu. The reasons were structural: public blockchains by design lack transaction privacy, prevent the central bank from controlling wallet balances, and resist regulation — properties that are features for cryptocurrency advocates but fatal flaws for a central bank instrument. This opened the Brazilian stablecoin market.

- Core incompatibilities: no transaction privacy, no balance privacy, and insufficient regulatory control — all inherent to public blockchain design.
 - Bitcoin's genesis block embeds the front page of a 2008 newspaper criticizing central banks, making the philosophical opposition to central banking a permanent, irremovable part of the protocol.
 - “Drex” unintentionally means “shit” (plural) in Hebrew (drek/drex), which caused public backlash from the Israeli community when the name was announced.
 - With Drex gone, Brazilian stablecoin developers can now compete in the market without facing government-backed competition; at least nine Brazilian Real-denominated stablecoins currently exist, including one by the instructor.
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12. BRICS Pay, Pix Internacional, and Global Stablecoin Trends

BRICS Pay was originally conceived in November 2019 at a BRICS Young Scientists Forum in Rio de Janeiro as a cryptocurrency to bypass SWIFT and US financial dominance, with an inflation/GDP-linked exchange rate. COVID and geopolitical disruptions delayed it, but it is now advancing. Simultaneously, Pix Internacional launched with Argentina, and Argentina's stablecoin ecosystem — driven by hyperinflation and exchange controls — is thriving.

- BRICS Pay was proposed by a researcher from the University of Kaliningrad (Russia) at the 2019 BRICS Young Scientists Forum in Rio de Janeiro; the instructor participated as one of two Brazilian representatives (cybersecurity team).
 - Original concept: a BRICS cryptocurrency with an exchange rate tied to the aggregate inflation and GDP growth of all BRICS nations, intended for import/export settlement to reduce SWIFT/USD dependence.
 - COVID (6 months after conception), then Russia's war, prevented full execution; a modified version is now moving forward — crypto rails for inter-country settlement, with citizen-facing interfaces resembling familiar payment apps.
 - Pix Internacional launched recently, beginning with Banco da Patagonia (Argentina); Brazilian and Argentine users can transact cross-border via Pix connected to the Central Bank infrastructure.
 - Argentina's stablecoin ecosystem is among the largest in Latin America: hyperinflation pushed citizens toward USDT/USDC via stablecoin swaps to obtain dollars without government exchange controls; Ethereum held its worldwide conference (Devcon) in Buenos Aires reflecting this prominence.
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13. Course Methodology & Project Orientation

The course follows a project-oriented methodology split into two phases: a taught, explanatory phase with interactive lectures, and a second phase centered on small-team student projects with partial deliveries. Individual projects are discouraged based on past experience showing they tend to fail.

- The course is project-oriented; students must put hands on to make things happen.
- The first part of the course is more lecture-based and explanatory; the second part focuses on student-led team projects with coding sessions.
- Individual projects are not allowed — only small team projects, as individual approaches have proven ineffective.
- The slide set serves as a narrative guide, but the actual lecture content is shaped dynamically by student interaction.

- The instructor intentionally does not over-prepare, as the value lies in live discussion and interaction.
 - Students are encouraged to re-raise questions in later, more appropriate lectures rather than expecting full answers immediately.
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14. Grading & Evaluation Structure

The semester grade is calculated from a combination of one theoretical exam and a series of practical partial deliveries. This distributed approach was deliberately chosen over a single final submission to prevent last-minute work and ensure ongoing engagement and quality.

- The university requires at least one exam and a final grade for each student; a single exam is considered unfair by the instructor.
 - Practical partial deliveries replace a single end-of-semester project submission; they allow for tweaking and course corrections if things go off track.
 - Deliverable deadlines are generally set **two weeks** after the relevant lecture, at **23:59 on a Wednesday**, and will be posted on Moodle.
 - The theoretical test is on **July 1**, conducted in a lab via Moodle Provas: approximately 12 questions, one hour, covering everything discussed throughout the semester.
 - Final projects must be submitted by **July 6** so the instructor can mark them by July 7.
 - The SINOVA competition finals are on **July 10**; two selected teams represent the course.
 - Attending an external competition earns **0.5 bonus points**; winning earns **1.0 bonus point**.
 - Recovery sessions are scheduled for **July 13**; final grades will be issued around **July 8**.
 - A final recovery activity exists but is rarely needed; students typically either pass or give up entirely.
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15. Course Deliverables and Final Pitch Format

The final assessment consists of a maximum five-minute slide pitch followed by a five-minute live or recorded demonstration of a working software artifact (MVP), uploaded to YouTube as a non-public video and submitted via link.

- The pitch is a slide presentation, no longer than five minutes.
- A software artifact (MVP) must be demonstrated in an additional five minutes.

- The demonstration is typically a video recorded and uploaded to YouTube (non-public), with the link submitted to the instructor.
 - The concept of MVP will be discussed in more detail later in the course.
 - The Canvas deliverable is expected around April 29 (two weeks after the April 15 lecture).
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16. Group Formation and Project Advice

Students are urged to form their groups on Moodle immediately, choose members wisely, and understand that grades are shared equally regardless of individual contribution — making team selection a critical decision.

- Groups can be created directly on Moodle; students can set a group name (permanent), description (editable), and a password for others to join — group names cannot be changed once set.
 - Grades for deliverables are automatically distributed to all group members — the instructor does not differentiate individual contributions.
 - The instructor candidly notes: typically one person works a lot, two observe, and one never shows up — and this is the student's responsibility to manage.
 - Group mixing during ecosystem events is encouraged to help students identify compatible teammates before committing.
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17. Upcoming Ecosystem Events

The course integrates real-world blockchain ecosystem events as part of the curriculum, with attendance counted and certificates available. The first is a Blockchain in Rio University Tour at AFE auditorium, and the second is an IP Village vibe coding session.

- The next session (replacing a regular lecture) is the **Blockchain in Rio University Tour Brazil** at the AFE auditorium — attendance will be checked.
 - Topics at the event: tokenization, digital assets, blockchain technology, crypto careers (Binance, Chainless, Bloxva, Blockchain in Rio); Binance typically gives out \$30 vouchers to audience members who answer questions; free pizza will be provided.
 - The **IP Village** event (April 1) will include a hands-on vibe coding challenge: students will work in teams to build software for a "city of the future."
 - A QR code will be available at events to register for a four-hour participation certificate applicable to the course.
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18. SINOVA Innovation Competition

UFSC's CInova secretariat runs a semester-end innovation competition across courses. The instructor selects two teams based on winning potential (not purely on quality), and the course has a strong winning record despite fierce competition from other departments.

- CInova (Secretaria de Inovação at UFSC) runs a competition among innovation and startup-focused courses; the course competes against other UFSC departments, including notably the Curitibaanos marijuana research lab and a capybara tracking device project.
 - The course has won approximately three times; selection of the two competing teams is based on the instructor's assessment of which teams are most likely to **win**, not necessarily the two best technical teams.
 - Last semester, the course placed third among 14–16 competing teams.
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19. Attendance & Student Engagement

Attendance is formally tracked due to university requirements, but it is not the primary basis for failure. However, since course discussions directly shape the deliverables, disengaged students naturally risk producing weaker project work. The course is explicitly designed for motivated, interested students.

- Attendance is recorded to satisfy university requirements, not to penalize students directly.
 - Students will not fail solely because of absences, but missed discussions can negatively affect deliverable quality.
 - The course is explicitly intended for interested students; those who are not engaged are not pressured to stay.
 - Discussions held during lectures directly influence and shape the project deliverables.
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Suggestions

- If you come from a non-technical background, focus first on understanding the *why* behind each cryptographic primitive (what problem it solves) rather than the mathematical detail — the conceptual logic is what drives design decisions in crypto startups.
- Start thinking early about an interdisciplinary problem in your own field that could benefit from immutability, decentralization, or programmable rules — the

most original startup ideas in this course have come from students outside computer science.

- Embrace AI-assisted coding tools from day one; treat them as a scaffold that lets you prototype and iterate quickly, but make sure you understand the blockchain logic your code is implementing.
 - When evaluating any blockchain project or use case, always ask: are all three pillars (integrity, rules, incentives) present and mutually reinforcing? If not, the system is fragile.
 - Form your project group on Moodle as soon as possible, set a meaningful group name carefully (it cannot be changed), and choose teammates wisely — grades are shared equally regardless of who does the work.
 - Attend the ecosystem events (Blockchain in Rio Tour, IP Village vibe coding session) not only for curriculum credit and certificates but also to network and identify potential reliable teammates before group formation is finalized.
 - Track the Moodle calendar regularly, as all schedule changes — including lecture cancellations due to travel, strikes, or other disruptions — will be updated there and not communicated through other channels in advance.
 - Treat the partial deliveries as genuine milestones rather than formalities — they exist to help you course-correct and improve incrementally rather than scrambling at the end.
 - Pay attention to the real-world policy landscape (Drex's cancellation, BRICS Pay, Pix Internacional, stablecoin proliferation) as these events directly shape where blockchain technology is heading and where entrepreneurial opportunities are opening up in Brazil and Latin America.
 - When studying smart contracts, internalize the core distinction from natural-language contracts: code-based rules are deterministic — you know the exact interpretation before execution, eliminating ambiguity-driven disputes.
 - Students should re-raise unanswered technical questions in specific later lectures designed for those topics rather than expecting full depth in introductory sessions — depth will come contextually.
 - Students with innovative project ideas should actively engage with guest speakers from Solana, Chainlink, and other ecosystems, as these contacts provide direct pathways to global hackathons (up to \$200,000 prizes), financing leads, and collaborative development opportunities.
 - For the SINOVA competition, focus on building a project with a strong, clear sales pitch and real technical credibility, as the instructor selects teams based on winning potential in a competitive field.
 - Aim to participate in external innovation competitions for bonus points — 0.5 for attending, 1.0 for winning — which can meaningfully impact your final grade.
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AI Suggestions

- The core of this lesson spans two interconnected themes: **'The Game-Theoretic Glue of Blockchain'** (how old cryptographic primitives are combined into a self-reinforcing, cheat-proof protocol) and **'The Architecture of Trust: from Centralized E-Money to Decentralized Blockchains and Smart Contracts'** (how blockchain progressively replaces reliance on central trusted parties with cryptographic primitives and consensus mechanisms). It's recommended to start with a mapping exercise — draw a diagram linking each building block (hash, public key, P2P, data structure) to the property it provides (integrity, identity, distribution, ordering) — and then run a side-by-side comparison exercise taking one traditional process (e.g., a rental agreement or bank transfer) and rewriting its rules first as a natural-language contract, then as pseudocode resembling a smart contract, to grasp deterministic, code-based trust through actively reconstructing why removing any single component breaks the system.
- Core content: Blockchain is not a new technology but a novel composition of 1950s–1990s components (linked lists → hash pointers; public-key crypto → wallets & signatures; P2P → distribution; game theory → incentive alignment) such that the protocol is a Nash equilibrium: honest participation dominates cheating, and cryptographic anchoring makes deviation provably costly. Each design choice (public vs. permissioned chain, anonymity vs. auditability) carries direct trade-offs that determine real-world applicability, as demonstrated by Drex's failure and Bitcoin's enduring success.
- A third pedagogical theme runs throughout: **'Distributed, Project-Based Evaluation as a Learning Strategy'** — rather than relying on a single high-stakes exam or last-minute project submission, learning is demonstrated progressively through small-team projects broken into timed partial deliveries, supplemented by one theoretical exam. Active participation in live discussions is essential, as these sessions directly shape deliverable expectations and allow for ongoing course correction. Start by mapping the semester's partial delivery schedule against your team's availability and project ideas, and identify what needs to be built and discussed in the first weeks of class.
- **Extracurricular Resources:**
 - [Bitcoin Whitepaper by Satoshi Nakamoto – the original document showing exactly how the components are stitched together:
<https://bitcoin.org/bitcoin.pdf>]
 - [Mastering Bitcoin by Andreas M. Antonopoulos (free on GitHub) – deep technical coverage of cryptographic primitives, wallets,

transactions, and consensus:

<https://github.com/bitcoinbook/bitcoinbook>]

- [MIT OpenCourseWare – Cryptography and Cryptanalysis (6.875) – covers hash functions, public-key crypto, commitments, and ZKPs at an accessible graduate level: <https://ocw.mit.edu/courses/6-875-cryptography-and-cryptanalysis-spring-2005/>]
- [a16z Crypto Canon – a curated reading list covering blockchain fundamentals, game theory, tokenomics, and ecosystem overviews, ideal for both technical and non-technical students: <https://a16zcrypto.com/posts/article/crypto-canon/>]
- [Ethereum Smart Contract documentation and Solidity introductory tutorials – directly relevant to the Solidity coding lecture and project deliverables: <https://docs.soliditylang.org/en/latest/introduction-to-smart-contracts.html>]
- [Binance Academy – Free beginner-to-advanced blockchain and crypto learning resources: <https://academy.binance.com>]
- [Business Model Canvas Guide (Strategyzer) – Learn how to complete a Canvas for your startup project: <https://www.strategyzer.com/canvas/business-model-canvas>]
- [Project-Based Learning: A Practical Guide (PBLWorks): <https://www.pblworks.org/what-is-pbl>]